DATA-550 PROJECT REPORT

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Abstract:

Diabetes is a growing issue in the United States with an alarming 49.6% of the adult population being diagnosed with either prediabetes or diabetes. To help combat this epidemic, we propose a machine learning model built on the Diabetes Health Indicators Dataset to better understand how we can accurately diagnose patients.

SPRING 2025
GRAND VIEW UNIVERSITY

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Introduction

Diabetes Is becoming a major issue in the United States with 38.4 million people, or 11.6% of the population, having diagnosed and undiagnosed diabetes[2]. We add another 97.6 million people 18+ years or older, or another 38% of the adult population, if we consider prediabetes[2]. These numbers are projected to increase unless something is done to change course.

In this report we will explore many characteristics of positively diagnosed diabetic and prediabetic individuals who took part in a yearly health risk survey. Our goal is to identify risk factors and preventative measures that can be mitigated or implemented into the daily routines of Americans to reverse the trend of diabetes diagnoses.

Data Description

This dataset was found on Kaggle[1]. The data is from The Behavioral Risk Factor Surveillance System (BRFSS) telephone survey conducted in 2015. It contains 253,680 rows and 22 columns. Each row represents an individual's response to the survey and each column represents the individuals' characteristics, habits, and biomarkers. There are no missing values in this dataset. Of note, this dataset is pre-cleaned.

Column name	Type	Description
Diabetes_012	Categorical	0 = no diabetes, 1 = prediabetes, 2 = diabetes
HighBP	Categorical	0 = no high BP, 1 = high BP
HighChol	Categorical	0 = no high cholesterol, 1 = high cholesterol
CholCheck	Categorical	0 = no cholesterol check in 5 years, 1 = yes cholesterol check in 5 years
BMI	Numerical	Body Mass Index
Smoker	Categorical	Have you smoked at least 100 cigarettes in your entire life, 0 = no 1 = yes
Stroke	Categorical	(Ever told) you had a stroke, 0 = no 1 = yes
HeartDiseaseorAttack	Categorical	coronary heart disease or myocardial infarction, 0 = no 1 = yes
PhysActivity	Categorical	physical activity in past 30 days - not including job 0 = no 1 = yes
Fruits	Categorical	Consume Fruit 1 or more times per day 0 = no 1 = yes
Veggies	Categorical	Consume Vegetables 1 or more times per day 0 = no 1 = yes

HvyAlcoholConsump	Categorical	Heavy drinkers (adult men having more than 14 drinks per week and adult women having more than 7 drinks per week) 0 = no 1 = yes
AnyHealthcare	Categorical	Have any kind of health care coverage, including health insurance, prepaid plans such as HMO, etc. 0 = no 1 = yes
NoDocbcCost	Categorical	Was there a time in the past 12 months when you needed to see a doctor but could not because of cost? 0 = no 1 = yes
GenHlth	Ordinal	Would you say that in general your health is: scale 1-5 1 = excellent 2 = very good 3 = good 4 = fair 5 = poor
MentHlth	Numerical	Now thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good? scale 1-30 days
PhysHlth	Numerical	Now thinking about your physical health, which includes physical illness and injury, for how many days during the past 30 days was your physical health not good? scale 1-30 days
DiffWalk	Categorical	Do you have serious difficulty walking or climbing stairs? 0 = no 1 = yes
Sex	Categorical	0 = female 1 = male
Age	Ordinal	13-level age category, 1 = 18-24, 9 = 60-64, 13 = 80 or older
Education	Ordinal	Education level scale 1-6, 1 = Never attended school or only kindergarten, 2 = Grades 1 through 8 (Elementary), 3 = Grades 9 through 11 (Some high school), 4 = Grade 12 or GED (High school graduate), 5 = College 1 year to 3 years (Some college or technical school), 6 = College 4 years or more (College graduate)
Income	Ordinal	Income scale 1-8, 1 = less than $10,000, 5 = less than $35,000, 8 = $75,000 or more

Questions of Interest

- What does the distribution of key health indicators (such as diabetes status, BMI, and general health) look like among survey respondents?
- Can we build a predictive model that accurately identifies individuals at risk for diabetes based on survey data?

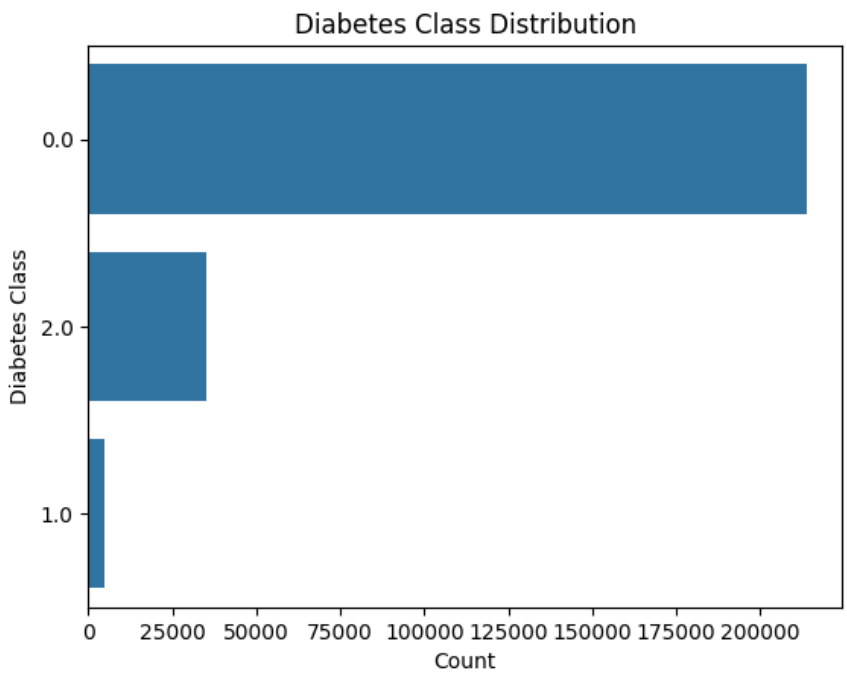
- Is it more effective to predict prediabetes and diabetes separately, or to group them into a single “at risk” category?

Exploratory Data Analysis

To better understand the data, we created a series of visualizations to explore the distribution of key health indicators, such as diabetes status, BMI, and general health. The goal of this analysis was to identify distinguishing characteristics of individuals with diabetes or prediabetes, with the aim of uncovering possible risk factors or preventive behaviors.

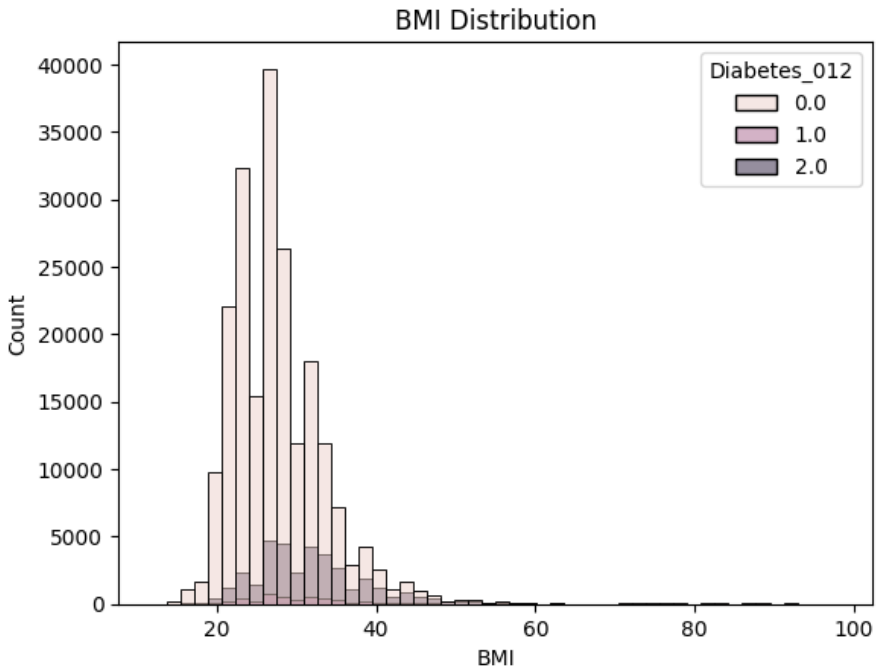
Figure 1: Diabetes Class Distribution Bar Chart

Commented [MM1]: Axis labels flipped



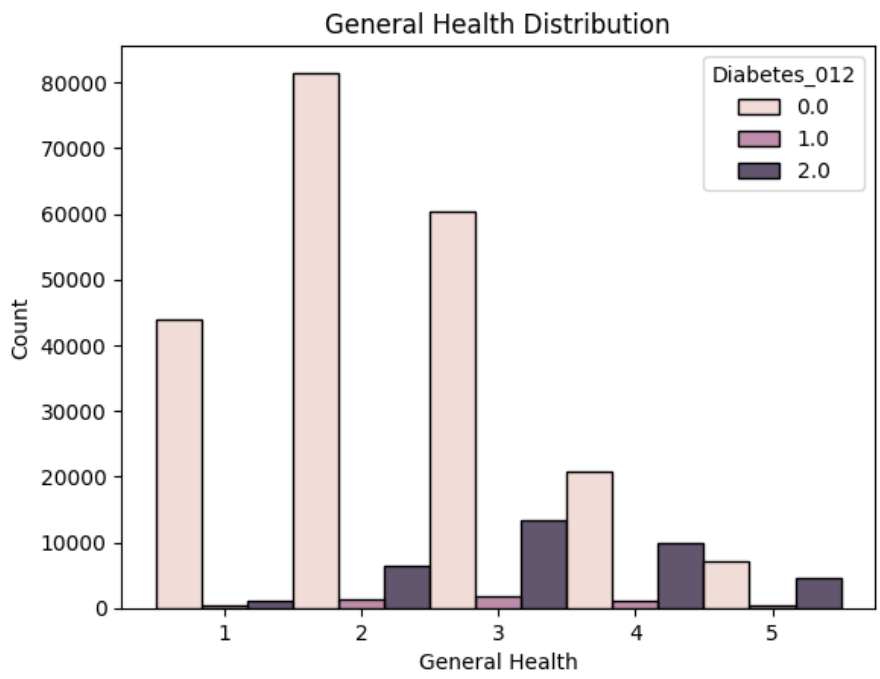
The chart above shows a clear class imbalance in our target variable: the majority of survey respondents reported no diabetes, while prediabetes and diabetes cases make up a much smaller portion of the data.

Figure 2: BMI Distribution Histogram



The histogram above shows a right-skewed BMI distribution: most respondents cluster between 20 and 40, with a long tail stretching toward higher values.

Figure 3: General Health Distribution Bar Chart



Commented [MM2]: Axis labels flipped

Commented [MM3R2]: Also would like general health to be in order (1,2,3,4,5)

From the above chart we can see most respondents ranked their own general health as very good, good, and excellent followed by fair and poor.

Modeling

For our baseline model, we decided to use a Random Forest Classifier and selected F1 score as our performance metric. As noted in our EDA, the dataset is highly imbalanced, with the "no diabetes" class comprising the majority of responses. Because of this, we avoided using accuracy—since a model could predict only "no diabetes" and still achieve a deceptively high score. Instead, we chose F1 score, which balances both precision and recall, providing a more informative measure when class distribution is uneven. Specifically, we use the macro-averaged F1 score, which calculates the F1 score independently for each class and then takes the unweighted mean, treating each class equally.

In our first run, we used Optuna to optimize the following hyperparameters between certain boundaries:

- N estimators: 100 to 300
- Max depth: 5 to 10
- Min samples split: 2 to 10
- Min samples leaf: 1 to 10

After 50 trials, using 5-fold cross validation, the best model had a macro F1 score of 36.8% on the train data set and a macro F1 score of 37.0% on the test set.

Figure 4: Model Hyperparameters

Parameters (6)

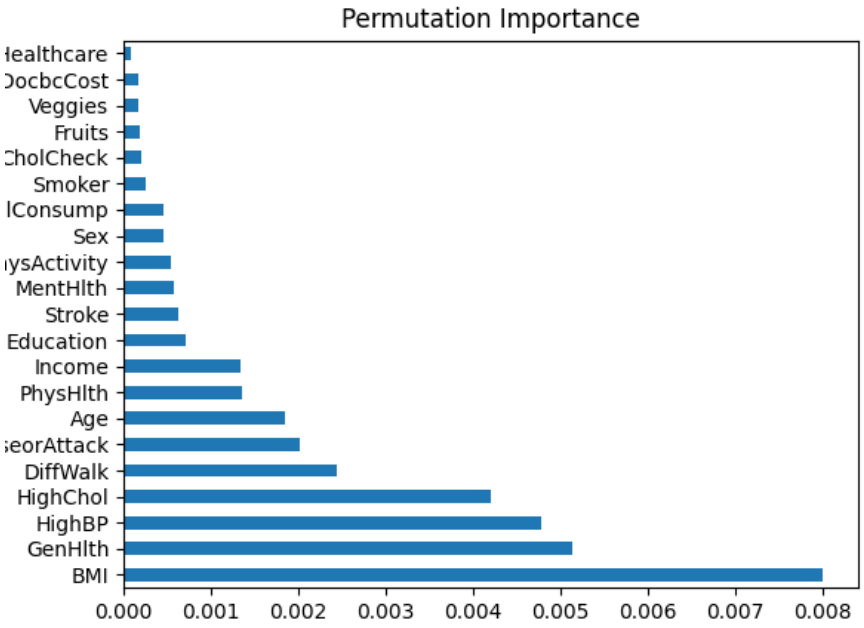
Search parameters

Parameter	Value
max_depth	10
min_samples_leaf	1
min_samples_split	10
n_estimators	101
n_jobs	-1
random_state	42

[1]

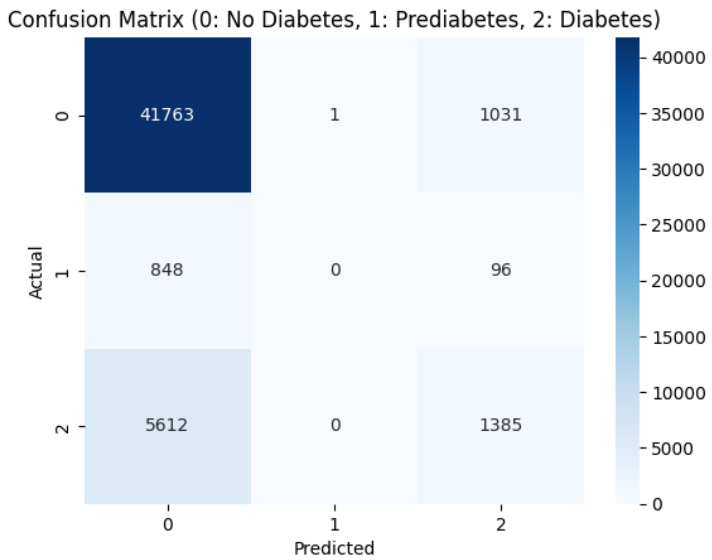
After our baseline Random Forest model, we experimented with a range of classification algorithms, including Extra Trees, Gradient Boosting, XGBoost, LightGBM, and CatBoost. We also explored feature engineering to improve performance, but these efforts did not lead to meaningful gains. Despite the variety of models tested, achieving a high F1 score remained difficult. The best observed macro-averaged F1 score hovered in the low 40s, highlighting the challenge posed by the imbalanced target variable and the overlap between diabetes and prediabetes cases.

Figure 5: Random Forest Permutation Importance



Although feature engineering didn't lead to improved performance, we used permutation importance to better understand which variables the model relied on most. This analysis confirmed the relevance of features like BMI, general health, and blood pressure. Still, performance remained limited — particularly in distinguishing between prediabetes and diabetes. This became clear when examining the confusion matrix, which showed that the model frequently misclassified prediabetes cases as diabetes, suggesting insufficient separation between those classes in the feature space.

Figure 6: Confusion Matrix



The confusion matrix confirmed the model's poor performance on the prediabetes class, with zero correct predictions. All prediabetes cases were misclassified as either no diabetes or diabetes, underscoring the model's inability to distinguish this class. This highlighted the lack of clear separation in the feature space and motivated us to simplify the task by combining diabetes and prediabetes into a single "at risk" category for binary classification.

After switching to binary classification, we saw a notable performance boost. With an XGBoost model, our F1 score rose from 37% in the baseline to 61%, marking a 24% improvement.

We achieved even better results by tuning the decision threshold between the "no diabetes" and "at risk" classes. Our best model, a LightGBM classifier with an optimized threshold of 0.35, reached an F1 score of 78.8%.

Run Name	Created	Duration	Source	Models	best_threshold	optuna_cv_f1	threshold_f1
CAT Optuna	37 minutes ago	32.6s	project_...	sklearn	0.339999999...	0.610042254...	0.769123288...
LGB Optuna	1 hour ago	22.3s	project_L...	sklearn	0.349999999...	0.613295350...	0.788472207...
XGB Optuna	1 hour ago	11.9s	project_...	sklearn	0.329999999...	0.612750301...	0.746695883...
GB Optuna	2 hours ago	1.4min	project_...	sklearn	0.359999999...	0.610628015...	0.777232434...
ET Optuna	3 hours ago	18.5s	project_...	sklearn	0.279999999...	0.519206802...	0.684372352...
RF Optuna	4 hours ago	55.0s	project_...	sklearn	0.289999999...	0.560118910...	0.698745797...

Figure 7: F1 scores and optimal thresholds

Conclusion & Further Study

In this paper, we explored the growing challenge of diabetes in the United States and developed predictive models to identify individuals at risk. Given the difficulty distinguishing between diabetes and prediabetes, we reframed the task as a binary classification problem—grouping the two conditions into a single “at risk” category. This restructuring proved effective: our optimized LightGBM model achieved an F1 score of 78.8%, demonstrating strong potential for real-world application. A model like this could support early intervention efforts, helping healthcare providers allocate resources more efficiently and encouraging at-risk individuals to seek care sooner.

From a clinical perspective, the binary framing was also appropriate. Although diabetes and prediabetes are distinct diagnoses, they share many overlapping risk factors and typically prompt similar interventions, such as lifestyle changes and increased monitoring. Combining them into a single predictive category allowed the model to deliver more consistent, actionable results without losing practical value.

Looking ahead, a promising direction for further study would be to explore automated machine learning (AutoML) frameworks such as Auto-sklearn, FLAML, or H2O AutoML. These tools could help identify better-performing models or hyperparameter settings by systematically searching a broader range of options than manual tuning allows. AutoML could also uncover alternative model architectures or preprocessing strategies that further improve performance beyond our current baseline.

References

Dataset:

https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset?select=diabetes_012_health_indicators_BRFSS2015.csv[1]

Research:

https://www.cdc.gov/diabetes/php/data-research/index.html[2]